C1 Notes: functions, graphs and equations

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This module is the grammar of upper-school mathematics. Calculus, exponentials and trigonometry are all statements about functions, so the time spent here on precise language pays off for the next four years.

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1. What a function is

A **function** (*functie*) f from a set A to a set B is a rule that gives **exactly one** output $f(x) \in B$ for every input $x \in A$.

- The set A of allowed inputs is the **domain** (*domein*), written D_f.
- The set of outputs that actually occur, $\{f(x) : x \in D_f\}$, is the **range** (*bereik*), written B_f.

The key words are "exactly one". The circle $x^2 + y^2 = 1$ is not the graph of a function of x, because $x = 0$ gives two values of y. The **vertical line test** says the same thing in pictures: a curve is the graph of a function of x if every vertical line meets it at most once.

In Dutch school mathematics, when only a formula is given, the domain is the **largest set of real numbers for which the formula makes sense**. Three things can go wrong:

1. dividing by zero;
2. a square root (or any even root) of a negative number;
3. a logarithm of a number that is not positive (this comes in C2).

Example 1. $f(x) = 1 + \sqrt{6 - 2x}$. Domain: $6 - 2x \geq 0$, so $x \leq 3$. $D_f = (-\infty, 3]$. Range: as x runs from 3 down to $-\infty$, $6 - 2x$ takes every value in $[0, \infty)$, so $\sqrt{6 - 2x}$ takes every value in $[0, \infty)$, and $f(x)$ every value in $[1, \infty)$. So $B_f = [1, \infty)$.

Note the argument for the range. It is not enough to say "the square root is at least 0, so $f(x) \geq 1$". That only shows the range is *inside* $[1, \infty)$. You must also show every value ≥ 1 is reached. Here that follows because $6 - 2x$ runs through all of $[0, \infty)$ and the square root is increasing and continuous.

Interval notation. $[a, b]$ includes both ends, (a, b) excludes them, and ∞ always gets a round bracket. Dutch books often write $\langle a, b \rangle$ for (a, b), so $\langle \leftarrow, 3]$ means $(-\infty, 3]$.

Even and odd functions

- f is **even** if $f(-x) = f(x)$ for all x in the domain. The graph is symmetric in the y-axis. Example: x^2, $x^4 + x^2$, $|x|$.
- f is **odd** if $f(-x) = -f(x)$. The graph is point-symmetric in the origin. Example: x^3, $\frac{1}{x}$, $x^3 - x$.

Most functions are neither: $x^2 + x$ at $x = 1$ gives 2 and at $x = -1$ gives 0.

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2. Standard functions

You should be able to sketch these instantly, with a few exact points, and state domain and range.

Function	Domain	Range	Notes
$ax + b$	$\mathbb{R}$	$\mathbb{R}$ if $a \neq 0$	straight line
x^2	$\mathbb{R}$	$[0, \infty)$	parabola, even
x^3	$\mathbb{R}$	$\mathbb{R}$	odd, increasing
x^n, n even	$\mathbb{R}$	$[0, \infty)$	flatter near 0, steeper beyond 1
$\sqrt{x}$	$[0, \infty)$	$[0, \infty)$	half a sideways parabola
$\sqrt[3]{x}$	$\mathbb{R}$	$\mathbb{R}$	inverse of x^3
$\frac{1}{x}$	$x \neq 0$	$y \neq 0$	hyperbola, asymptotes $x = 0$, $y = 0$
$\frac{1}{x^2}$	$x \neq 0$	$(0, \infty)$	even
$\lvert x \rvert$	$\mathbb{R}$	$[0, \infty)$	V-shape

Power functions with negative and fractional exponents are linked by the power rules from stage B: $x^{-n} = \frac{1}{x^n}$ and $x^{1/n} = \sqrt[n]{x}$, $x^{p/q} = \sqrt[q]{x^p}$.

All of $x^2, x^4, x^6, \ldots$ pass through $(0, 0)$, $(1, 1)$ and $(-1, 1)$. Between -1 and 1 a higher power is *closer* to zero ($0.5^4 < 0.5^2$), beyond 1 it is larger. The same holds for odd powers through $(-1, -1)$, $(0, 0)$, $(1, 1)$.

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3. Transformations

Start from the graph of $y = f(x)$. There are two kinds of changes: those **outside** the function (they act on y) and those **inside** (they act on x).

New formula	Effect on the graph	Dutch phrase
$y = f(x) + q$	translate q up	translatie $(0, q)$
$y = f(x - p)$	translate p to the **right**	translatie $(p, 0)$
$y = a\, f(x)$	stretch by factor a from the x-axis (vertical)	vermenigvuldiging t.o.v. de x-as met a
$y = f(x/b)$	stretch by factor b from the y-axis (horizontal)	vermenigvuldiging t.o.v. de y-as met b
$y = -f(x)$	reflect in the x-axis	spiegeling in de x-as
$y = f(-x)$	reflect in the y-axis	spiegeling in de y-as

Why does $f(x - p)$ move the graph to the right?

The point $(a, f(a))$ lies on the old graph. On the new graph $y = f(x - p)$, which x gives the same height $f(a)$? We need $x - p = a$, so $x = a + p$. Every height is now reached p units further to the right. The inside changes always act "backwards": to stretch horizontally by factor b you *divide* x by b, because the new graph must reach at $x = ab$ the height the old one reached at a.

A good way to avoid mistakes: a transformation maps each point (x, y) of the old graph to a new point. For a translation (p, q) that is $(x, y) \mapsto (x + p, y + q)$. For a stretch from the y-axis by b it is $(x, y) \mapsto (bx, y)$. Write the old point as $(u, f(u))$, apply the map, and eliminate u.

Order matters

Example 2. Start with $y = \sqrt{x}$. Apply in this order: (1) translate $(-3, 0)$; (2) stretch from the x-axis by factor -2 (this is a stretch by 2 plus a reflection); (3) translate $(0, 5)$.

1. $y = \sqrt{x+3}$
2. $y = -2\sqrt{x+3}$
3. $y = -2\sqrt{x+3}+5$

If you translate up by 5 *before* stretching, you get $y = -2(\sqrt{x+3}+5) = -2\sqrt{x+3}-10$, a different graph. The rule: each step acts on the whole formula you have so far. Outside steps wrap around the outside; inside steps replace every x.

Example 3. $f(x) = x^2 - 2x$. The graph of g arises from that of f by a stretch from the y-axis by factor 2, followed by a translation $(0, 3)$. Then

$$g(x) = f\left(\tfrac{x}{2}\right) + 3 = \tfrac{x^2}{4} - x + 3.$$

Check with the vertex: f has vertex $(1, -1)$; stretching horizontally by 2 gives $(2, -1)$; translating gives $(2, 2)$. Indeed $g(2) = 1 - 2 + 3 = 2$.

> **Going further: transformations as composition.** A transformation of the graph is a composition (*samenstelling*) of functions. $y = a\,f(x-p)+q$ is $x \mapsto x-p \mapsto f(x-p) \mapsto af(x-p) \mapsto af(x-p)+q$. The inside steps happen *before* f, the outside steps *after*. This view explains why inside changes are "backwards": to undo the order you must invert them.

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4. Inverse functions

If f takes x to y, the **inverse function** (*inverse functie*) f^{-1} takes y back to x:

$$f^{-1}(f(x)) = x \quad \text{for all } x \in D_f, \qquad f(f^{-1}(y)) = y \quad \text{for all } y \in B_f.$$

This only works if no two inputs give the same output, that is if f is **one-to-one** (*injectief*): $f(a) = f(b) \Rightarrow a = b$. Every strictly increasing or strictly decreasing function is one-to-one. Graphically: every **horizontal** line meets the graph at most once.

Then $D_{f^{-1}} = B_f$ and $B_{f^{-1}} = D_f$.

Method. Write $y = f(x)$, solve for x in terms of y, then rename the variables.

Example 4. $f(x) = \dfrac{2x+1}{x-3}$, $x \neq 3$.

$$y(x-3) = 2x+1 \;\Rightarrow\; yx - 2x = 3y + 1 \;\Rightarrow\; x(y-2) = 3y+1 \;\Rightarrow\; x = \frac{3y+1}{y-2}.$$

So $f^{-1}(x) = \dfrac{3x+1}{x-2}$, with domain $x \neq 2$. This also shows that the range of f is all $y \neq 2$: the value $y = 2$ is the only one we cannot solve for.

Example 5 (restricting the domain). $f(x) = x^2 - 4x + 1$ is not one-to-one on $\mathbb{R}$, because a parabola is symmetric. Completing the square, $f(x) = (x-2)^2 - 3$. On the domain $x \geq 2$ it is increasing, so it has an inverse there:

$$y = (x-2)^2 - 3 \Rightarrow (x-2)^2 = y+3 \Rightarrow x - 2 = \sqrt{y+3}$$

(the positive root, because $x \geq 2$). So $f^{-1}(x) = 2 + \sqrt{x+3}$ with domain $x \geq -3$.

Why the graph is reflected in $y = x$

The point (a, b) lies on the graph of f exactly when $f(a) = b$, which is exactly when $f^{-1}(b) = a$, which is exactly when (b, a) lies on the graph of f^{-1}. Swapping the coordinates of a point is the reflection in the line $y = x$ (the line $y = x$ is the perpendicular bisector of the segment from (a, b) to (b, a)). So the whole graph is reflected.

> **Common mistake.** $f^{-1}(x)$ is not $\dfrac{1}{f(x)}$. The notation f^{-1} means "the inverse under composition", not "one over".
>
> **Going further: intersections of f and f^{-1}.** If f is *increasing*, the graphs of f and f^{-1} can only meet on the line $y = x$. Proof: suppose they meet at (a, b), so $f(a) = b$ and $f^{-1}(a) = b$, i.e. $f(b) = a$. If $a < b$ then, since f is increasing, $f(a) < f(b)$, i.e. $b < a$, a contradiction. Similarly $a > b$ is impossible. So $a = b$. This is a very useful shortcut on exams: to intersect an increasing f with its inverse, solve $f(x) = x$. For *decreasing* functions it fails: $f(x) = -x$ is its own inverse, so the graphs meet everywhere.

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5. Equations and inequalities

Root equations

To solve $\sqrt{A} = B$ you square both sides: $A = B^2$. But squaring is **not reversible**: $B^2 = A$ is also true when $\sqrt{A} = -B$. So squaring can introduce **false solutions** (*schijnoplossingen*). You must always check the solutions in the original equation (or require $B \geq 0$ in advance).

Example 6. Solve $\sqrt{x+7} = x + 1$. Squaring: $x + 7 = x^2 + 2x + 1$, so $x^2 + x - 6 = 0$, $(x + 3)(x - 2) = 0$, $x = -3$ or $x = 2$. Check $x = -3$: left $\sqrt{4} = 2$, right -2. False solution. Check $x = 2$: left 3, right 3. Correct. **Answer:** $x = 2$.

Graphically: the line $y = x + 1$ meets the half-parabola $y = \sqrt{x+7}$ once; the false solution is where it meets the *lower* half $y = -\sqrt{x+7}$, which the squaring step silently included.

Rational equations

$\dfrac{A}{B} = 0 \iff A = 0$ and $B \neq 0$. And $\dfrac{A}{B} = \dfrac{C}{D}$ can be cross-multiplied to $AD = BC$, provided $B \neq 0$ and $D \neq 0$: again check at the end.

Inequalities: the method that always works

To solve $f(x) > g(x)$ (in any form):

1. Solve the **equation** $f(x) = g(x)$ exactly.
2. Find the points where f or g is not defined or jumps (vertical asymptotes, domain boundaries).
3. These points cut the line into intervals. On each interval the sign of $f - g$ is constant (for the functions in this module, which are continuous on each interval). Test one point per interval, or use a sketch.
4. Write the answer, taking care with endpoints.

This is the **sign diagram** (*tekenschema*) method. The exam phrase is often "los exact op" for an inequality: you solve the equation exactly and then *use the graph* (sketch or GR screen) to read off the intervals. Write both steps down.

Example 7. Solve $\dfrac{x+2}{x-1} \geq 3$. Do **not** multiply by $x-1$: you don't know its sign. Instead bring everything to one side:

$$\frac{x+2}{x-1} - 3 = \frac{x+2-3(x-1)}{x-1} = \frac{5-2x}{x-1} \geq 0.$$

Zero of the numerator: $x = \frac{5}{2}$. Zero of the denominator: $x = 1$.

x	$x<1$	1	$1<x<\frac{5}{2}$	$\frac{5}{2}$	$x>\frac{5}{2}$
$5-2x$	$+$		$+$	0	$-$
$x-1$	$-$	0	$+$		$+$
quotient	$-$	undefined	$+$	0	$-$

Answer: $1 < x \leq \frac{5}{2}$.

Example 8 (polynomial). $x^3 > 4x$. Then $x(x-2)(x+2) > 0$. Zeros $-2, 0, 2$; the sign alternates (each factor appears once), and for large x the expression is positive. So the solution is $-2 < x < 0$ or $x > 2$.

> **Common mistake.** Dividing $x^3 > 4x$ by x to get $x^2 > 4$. That loses $-2 < x < 0$ and is wrong for negative x, where the inequality sign must flip.

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6. Asymptotes and limit behaviour

Limit notation

$\lim\limits_{x\to\infty} f(x) = L$ means: $f(x)$ gets as close to L as you like when x is large enough. We treat limits informally in this module and more carefully in C3.

Vertical asymptotes

A rational function $\dfrac{P(x)}{Q(x)}$ has a **vertical asymptote** $x = a$ when $Q(a) = 0$ and $P(a) \neq 0$. Near a the denominator is tiny and the numerator is not, so $|f(x)|$ becomes huge.

If $P(a) = 0$ *and* $Q(a) = 0$, then $x - a$ is a factor of both. Cancel it. If after cancelling the denominator is no longer zero at a, the graph has a **hole** (*perforatie*): a single missing point.

Example 9. $f(x) = \dfrac{x^2-4}{x^2-x-2} = \dfrac{(x-2)(x+2)}{(x-2)(x+1)}$. For $x \neq 2$ this equals $\dfrac{x+2}{x+1}$. So:

- vertical asymptote $x = -1$;
- hole at $x = 2$: the point $\left(2, \frac{4}{3}\right)$ is missing from the graph;
- horizontal asymptote $y = 1$ (see below).

Horizontal asymptotes

For $x \to \pm\infty$ compare the degrees of numerator and denominator. The trick is to divide numerator and denominator by the highest power of x in the denominator.

Example 10. $f(x) = \dfrac{3x^2-1}{x^2+4} = \dfrac{3-\frac{1}{x^2}}{1+\frac{4}{x^2}} \to \dfrac{3-0}{1+0} = 3$ as $x \to \pm\infty$. Horizontal asymptote $y = 3$.

A cleaner way to see this, and the way to get the range: write

$$f(x) = \frac{3(x^2+4)-13}{x^2+4} = 3 - \frac{13}{x^2+4}.$$

Now $x^2+4 \geq 4$, so $0 < \frac{13}{x^2+4} \leq \frac{13}{4}$, with the maximum at $x=0$. Therefore $-\frac{1}{4} \leq f(x) < 3$, and every value in between occurs. Range $\left[-\frac{1}{4}, 3\right)$. The function never reaches its asymptote.

> **Common mistake.** "A graph never crosses its asymptote." False for horizontal asymptotes: $\dfrac{x}{x^2+1}$ has asymptote $y=0$ and crosses it at $x=0$. An asymptote describes behaviour far away, not nearby.

Summary for $\dfrac{P}{Q}$ with degrees p and q:

- $p < q$: horizontal asymptote $y = 0$;
- $p = q$: horizontal asymptote $y = \dfrac{\text{leading coefficient of } P}{\text{leading coefficient of } Q}$;
- $p = q+1$: **oblique asymptote** (*scheve asymptoot*), found by long division;
- $p > q+1$: no straight-line asymptote.

Example 11 (oblique asymptote). $f(x) = \dfrac{2x^2+3x}{x-1}$. Long division (or writing $2x^2+3x = (x-1)(2x+5)+5$) gives

$$f(x) = 2x + 5 + \frac{5}{x-1}.$$

As $x \to \pm\infty$, $\frac{5}{x-1} \to 0$, so the graph approaches the line $y = 2x+5$. Vertical asymptote $x = 1$.

Root functions

$\sqrt{x}$ has no horizontal asymptote ($\sqrt{x} \to \infty$, slowly). But $\sqrt{x^2+1} - x$ does: multiply by the conjugate,

$$\sqrt{x^2+1} - x = \frac{(x^2+1)-x^2}{\sqrt{x^2+1}+x} = \frac{1}{\sqrt{x^2+1}+x} \to 0 \quad (x \to \infty).$$

The conjugate trick will return in C3 when we differentiate $\sqrt{x}$ from the definition.

A modelling example

Example 12. The number of visitors to a website after the launch of a new game is modelled by $N(t) = \dfrac{800t}{t+4}$, t in days, $t \geq 0$.

a. Which number of visitors does N approach in the long run? Dividing by t: $N = \dfrac{800}{1+4/t} \to 800$.

b. Bereken exact from which day on $N > 600$. Since $t+4 > 0$ we may multiply: $800t > 600t + 2400$, so $t > 12$.

c. Write t as a function of N. $N(t+4) = 800t \Rightarrow t(800-N) = 4N \Rightarrow t = \dfrac{4N}{800-N}$, for $0 \leq N < 800$.

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7. Absolute value

$$|x| = \begin{cases} x & x \geq 0 \\ -x & x < 0 \end{cases}$$

Geometrically, $|a-b|$ is the distance between a and b on the number line. This is the most useful way to think about it:

- $|x-3| = 5$: the points at distance 5 from 3, so $x = -2$ or $x = 8$.
- $|x-3| < 5$: the points closer than 5 to 3, so $-2 < x < 8$.
- $|x-3| > 5$: $x < -2$ or $x > 8$.

For more complicated equations, **split into cases** according to the sign of what is inside the absolute value bars.

Example 13. Solve $|x^2 - 4| = 3x$.

Case 1: $x^2 - 4 \geq 0$ (that is $x \leq -2$ or $x \geq 2$). Then $x^2 - 4 = 3x$, $x^2 - 3x - 4 = 0$, $x = 4$ or $x = -1$. Only $x = 4$ satisfies the case condition.

Case 2: $x^2 - 4 < 0$ ($-2 < x < 2$). Then $4 - x^2 = 3x$, $x^2 + 3x - 4 = 0$, $x = 1$ or $x = -4$. Only $x = 1$ satisfies the case condition.

Check: $|16 - 4| = 12 = 3 \cdot 4$ and $|1 - 4| = 3 = 3 \cdot 1$. **Answer:** $x = 1$ or $x = 4$.

Graphs. The graph of $y = |f(x)|$ is the graph of f with every part below the x-axis reflected upwards. The graph of $y = f(|x|)$ is the right half of f's graph together with its mirror image in the y-axis (it is always even).

> **Going further: $\sqrt{x^2} = |x|$.** Not x. For $x = -3$: $\sqrt{9} = 3 = |-3|$. Many errors in root equations come from "simplifying" $\sqrt{x^2}$ to x.

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8. Writing CE answers

Dutch exam markers follow a *correctievoorschrift* that awards points for intermediate steps. Some habits that earn the points:

- **"Bereken exact"**: no decimal approximations at any stage; answers like $\frac{5}{2}$ or $2 + \sqrt{3}$.
- **"Toon aan" / "Bewijs"**: start from what is given, end with exactly the statement asked, and say so ("dus …").
- **Inequalities**: write the equation you solve, its solutions, and then a sentence or sketch showing how you read off the intervals.
- **Using the GR**: name what you entered ("intersect van $y_1 = \ldots$ en $y_2 = \ldots$ geeft $x \approx \ldots$") and round only at the end.

A complete CE-style worked example

Example 14. Given is $f(x) = 2 - \sqrt{4 - x}$.

a. Give the domain and range of f. $4 - x \geq 0$ so $D_f = (-\infty, 4]$. $\sqrt{4-x}$ takes all values in $[0, \infty)$, so f takes all values in $(-\infty, 2]$. $B_f = (-\infty, 2]$.

b. Toon aan dat $f^{-1}(x) = 4 - (2-x)^2$ with $x \leq 2$. From $y = 2 - \sqrt{4-x}$: $\sqrt{4-x} = 2 - y$, which requires $y \leq 2$. Squaring: $4 - x = (2-y)^2$, so $x = 4 - (2-y)^2$. Swapping the letters gives $f^{-1}(x) = 4 - (2-x)^2$ with domain $B_f = (-\infty, 2]$.

c. Bereken exact the coordinates of the intersection points of the graphs of f and f^{-1}. f is increasing (as x increases, $4-x$ decreases, so $\sqrt{4-x}$ decreases and f increases). So intersection points lie on $y = x$ (see §4). Solve $2-\sqrt{4-x} = x$: $\sqrt{4-x} = 2-x$, square: $4-x = 4-4x+x^2$, so $x^2 - 3x = 0$, $x = 0$ or $x = 3$. Check: $x = 3$ gives $\sqrt{1} = 1 \neq -1$, false. $x = 0$ gives $2 = 2$. The only intersection point is $(0, 0)$.

Summary

- A function gives exactly one output per input; domain and range must be justified, not guessed.
- Inside changes act on x "backwards"; outside changes act on y directly; order matters.
- Inverse = swap x and y; exists when one-to-one; graph reflected in $y = x$.
- Squaring can create false solutions: always check.
- Inequalities: solve the equation, then use a sign diagram or a graph.
- Asymptotes: denominator zero (vertical), degree comparison (horizontal), long division (oblique).
- Absolute value = distance; split into cases.